# SHRI VIDHYABHARATHI MATRIC HR.SEC.SCHOOL

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## COMMON PUBLIC EXAMINATION –MARCH –2024

**XII - MATHEMATICS** **TENTATIVE ANSWER KEY**

| PART – I | | | |
|---|---|---|---|
| **Q.No** | **CODE – A** | **CODE – B** | **MARKS** |
| 1. | (a) $\frac{8}{3}$ | (a) 45° | 1 |
| 2. | (d) $\frac{3\pi a^4}{16}$ | (d) 8 | 1 |
| 3. | (a) 10 | (a) $\frac{8}{3}$ | 1 |
| 4. | (d) 2 | (a) 10 | 1 |
| 5. | (d) 8 | (d) $\frac{1}{\sqrt{5}}$ | 1 |
| 6. | (a) $[\vec{\alpha}, \vec{\beta}, \vec{\gamma}] = 0$ | (c)-4 | 1 |
| 7. | (d) $\frac{d^2y}{dx^2}$ -y=0 | (a) (b) $\frac{-q}{r}$ | 1 |
| 8. | (b) y =0 | (d) $\frac{3\pi a^4}{16}$ | 1 |
| 9. | (d) $\frac{1}{(x+1)^2}$ dx | (c) (d) adj(AB)=(adjA) (adjB) | 1 |
| 10. | (a) $x^2+y^2$ | (a) (a) $[\vec{\alpha}, \vec{\beta}, \vec{\gamma}] = 0$ | 1 |
| 11. | (c) 2 | (c) (d) $\frac{d^2y}{dx^2}$ -y=0 | 1 |
| 12. | (d) parabola | (c) 2 | 1 |
| 13. | (c) $\begin{bmatrix} 5 & -2 \\ 3 & -1 \end{bmatrix}$ | (d) parabola | 1 |
| 14. | (a) 45° | (d) 2 | 1 |
| 15. | (d) $\frac{1}{\sqrt{5}}$ | (c) $\begin{bmatrix} 5 & -2 \\ 3 & -1 \end{bmatrix}$ | 1 |
| 16. | (b) $\frac{-q}{r}$ | (a) $x^2+y^2$ | 1 |
| 17. | (a)0 | (b) y =0 | 1 |
| 18. | (c) 2 | (a)0 | 1 |
| 19. | (d) adj(AB)=(adjA) (adjB) | (c) 2 | 1 |
| 20. | (c) -4 | (d) $\frac{1}{(x+1)^2}$ dx | 1 |



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| | | |
|---|---|---|
| | **PART – II** | |
| 21. | $\sum_{n=1}^{12} i^n = 0$ | 2 |
| 22. | $\alpha^2 + \beta^2 = \frac{-3}{4}$ ; $\alpha^2\beta^2 = \frac{169}{4}$<br>$4x^2 + 3x + 169 = 0$ | 1<br>1 |
| 23. | df=$(2x+3)$dx<br>=0.18 | 1<br>1 |
| 24. | y=mx ; $\frac{dy}{dx}$=m<br>y = x$\frac{dy}{dx}$ | 1<br>1 |
| 25. | mean $= \int_1^2 2x(x-1)dx = \frac{5}{3}$ | 2 |
| 26. | $(x+3)^2+(y+4)^2=9$<br>$x^2 + y^2 + 6x + 8y + 16 = 0$ | 1<br>1 |
| 27. | $\begin{vmatrix} -1 & 3 \\ 4 & -7 \end{vmatrix} = -5$<br>ρ(A)=2 | 1<br>1 |
| 28. | $\int_0^{\frac{\pi}{2}} \sin^{10} x dx = \frac{63\pi}{512}$ | 2 |
| 29. | $\lim_{x\to1} \frac{x^2-3x+2}{x^2-4x+3} = \lim_{x\to1} \frac{2x-3}{2x-4}$<br>$= \frac{1}{2}$ | 1<br>1 |
| 30. | $\begin{vmatrix} 2 & -1 & 3 \\ 1 & -1 & 0 \\ 3 & -1 & 6 \end{vmatrix} = 0$<br>∴The given vectors are coplanar | 1<br>1 |
| | **PART – III** | |
| 31. | Let, cot$\boldsymbol{\alpha} = \frac{1}{\sqrt{x^2-1}}$<br>sec$\boldsymbol{\alpha}$=x ; $\boldsymbol{\alpha}$=sec$^{-1}$x<br>cot$^{-1}\left(\frac{1}{\sqrt{x^2-1}}\right)$= sec$^{-1}$x | 1<br>1<br>1 |
| 32. | $y = \frac{-1}{4}(x^2 + 6x + 5)$<br>$\frac{dy}{dx} = \frac{-1}{4}(2x+6)$<br>m=-2<br>Equation of tangent:2x+y+1=0<br>Equation of normal: x-2y-7=0 | 1<br>1<br>1 |



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| | | |
|---|---|---|
| 33. | LHS= $\begin{vmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ -1 & 0 & 1 \end{vmatrix}$ $[\vec{a},\vec{b},\vec{c}]=0$<br>$\therefore[\vec{a}-\vec{b},\vec{b}-\vec{c},\vec{c}-\vec{a},]=0$ | 2<br>1 |
| 34. | $u(\lambda x,\lambda y) = \frac{\lambda^{2(x^2+y^2)}}{\sqrt{\lambda}\sqrt{x+y}} = \lambda^{\frac{3}{2}}u(x,y)$<br>$n=\frac{3}{2}$<br>By euler's theorem: $x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}=\frac{3}{2}u$ | 1<br>2 |
| 35. | x+y=12⟹y=12-x<br>p=xy =x(12-x)<br>p'(x)=12-2x⟹$p'(x)=0$<br>x=6<br>p''(x) = -2<0, p is maximum.<br>∴x=6,y=6 | 1<br>1<br>1 |
| 36. | $I=\int_{\frac{\pi}{8}}^{\frac{3\pi}{8}} \frac{1}{1+\sqrt{\tan x}}\,dx$--------1<br>$I=\int_{\frac{\pi}{8}}^{\frac{3\pi}{8}} \frac{\sqrt{\tan x}}{1+\sqrt{\tan x}}\,dx$--------2<br>From 1+2<br>$I=\frac{\pi}{8}$ | 1<br>2 |
| 37. | $\left(\frac{1+i}{1-i}\right)^3-\left(\frac{1-i}{1+i}\right)^3$ = -i-i<br>= -2i | 2<br>1 |
| 38. | $\int\frac{dy}{1+y^2}=\int\frac{dx}{1+x^2}$<br>$\tan^{-1}y=\tan^{-1}x+c$ | 1<br>2 |
| 39. | x: 0, 1, 2, 3<br>f(x): $\frac{1}{8}$, $\frac{3}{8}$, $\frac{3}{8}$, $\frac{1}{8}$ | 3 |
| 40. | \|A\| = - 1<br>$\|adj(adjA)\|=\|A\|^{(n-1)^2}=(-1)^4=1$<br>**Note: Any other method** | 1<br>2 |
| | **PART – IV** | |
| 41.(a) | Rough Diagram<br>Point of intersection=$\left(\frac{3}{2},\frac{9}{4}\right)$<br>$m_1=3$ ; $m_2=-3$<br>$\theta=\tan^{-1}\left(\frac{3}{4}\right)$ | 1<br>1<br>1<br>2 |



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**41.(b)**

$$\tan^{-1}\left(\frac{\frac{x-1}{x-2}+\frac{x+1}{x+2}}{1-\left(\frac{x-1}{x-2}\right)\left(\frac{x+1}{x+2}\right)}\right)=\frac{\pi}{4}$$ — 1

$$\frac{(x-1)(x+2)+(x+1)(x-2)}{(x-2)(x+2)+(x+1)(x-1}=1$$ — 2

$$2x^2-4=-3 \Longrightarrow X=\pm\frac{1}{\sqrt{2}}$$ — 2

**42.(a)**

| x | 2 | 4 | 6 | 8 | 10 |
|---|---|---|---|---|---|
| f(x) | $\frac{1}{36}$ | $\frac{4}{36}$ | $\frac{10}{36}$ | $\frac{12}{36}$ | $\frac{9}{36}$ |

— 2

$$F(x)=\begin{cases} 0 \text{ , if } -\infty < x < 2 \\ \frac{1}{36} \quad \text{if } 2 \le x < 4 \\ \frac{5}{36} \quad \text{if } 4 \le x < 6 \\ \frac{15}{36} \quad \text{if } 6 \le x < 8 \\ \frac{27}{36} \quad \text{if } 8 \le x < 10 \\ 1 \text{ if } 10 \le x < \infty \end{cases}$$ — 2

$P(4 \le x < 10)=\frac{13}{18}$ — 1

**42.(b)**

$$\left(\frac{2z+1}{iz+1}\right)=\frac{2(x+iy)+1}{i(x+iy)+1}$$ — 1

$$\text{Im}\left(\frac{(2x+1)+i2y}{(1-y)+ix}\right)=0$$ — 2

$$\Rightarrow\frac{2y(1-y)-x(2x+1)}{(1-y)^2+x^2}=0$$ — 1

$2x^2+2y^2+x-2y=0.$ — 1

**43.(a)**

Rough Diagram — 1

$r=\frac{5}{12}h$ — 1

$V=\frac{1}{3}\pi\left(\frac{5}{12}h\right)\left(\frac{5}{12}h\right)h$ — 1

$\frac{dV}{dt}=\frac{25}{144}\pi\times h^2\times\frac{dh}{dt}$ — 1

$\frac{dh}{dt}=\frac{9}{10\pi}$ m/min — 1

**43.(b)**

Diagram — 1

$\overrightarrow{OP}=\cos\alpha\vec{i}+\sin\alpha\vec{j}$

$\overrightarrow{OQ}=\cos\beta\vec{j}+\sin\beta\vec{j}$ — 1

$$\overrightarrow{OQ}\times\overrightarrow{OP}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ \cos\beta & \sin\beta & 0\\ \cos\alpha & \sin\alpha & 0\end{vmatrix}$$

$=\hat{k}(\sin\alpha\ \cos\beta-\cos\alpha\sin\beta)$ . . . . .(1) — 1

By definition

$\overrightarrow{OQ}\times\overrightarrow{OP}=|\overrightarrow{OQ}||\overrightarrow{OP}|\sin(\alpha-\beta)\,\hat{k}=\sin(\alpha-\beta)\,\hat{k}$. . . . .(2) — 1

From (1) and (2)

$\sin(\alpha-\beta)=\sin\alpha\ \cos\beta-\cos\alpha\sin\beta$ — 1



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| | | |
|---|---|---|
| 44.(a) | Rough Diagram<br>$x^2 = -4ay$<br>$a = \frac{9}{10}$<br>It's passes through,$(x_1,-7.5)$<br>$x^2 = -4\left(\frac{9}{10}\right)y$<br>$x_1 = 3\sqrt{3}$mt. | 1<br>1<br>1<br><br>1<br>1 |
| 44.(b) | $P = \frac{1}{x}$, $Q = \sin x$<br>I.F= $e^{\int p\,dx} = x$<br>$xy = \int x\sin x\,dx + c$<br>$xy = \sin x - x\cos x + c$ | 1<br>1<br>2<br>1 |
| 45.(a) | $\frac{dx}{dt}\ \alpha\ x \Rightarrow \frac{dx}{dt} = kx \Rightarrow x = ce^{kt}$<br>$t=0$ ; $X=x_0 \Rightarrow C=X_0$<br>$X=X_0e^{kt}$<br>$t=5$ ; $X=3x_0 \Rightarrow e^{5K}=3$<br>$t=10$ ; $X=9x_0$ | 1<br>1<br>1<br>1<br>1 |
| 45.(b) | $\vec{a}=2\hat{i}+2\hat{j}+\hat{k}$ ; $\vec{b}=2\hat{i}+3\hat{j}+3\hat{k}$ ; $\vec{c}=3\hat{i}+2\hat{j}+\hat{k}$<br>**Parametric form**<br>$\vec{r}=(2\hat{i}+2\hat{j}+\hat{k})+s(2\hat{i}+3\hat{j}+3\hat{k})+t(3\hat{i}+2\hat{j}+\hat{k})$<br>**(or) non Parametric form**<br>$\vec{r}.(3\hat{i}-7\hat{j}+5\hat{k})=-3$<br>**Cartesian form**<br>$\begin{vmatrix} x-2 & y-2 & z-1 \\ 2 & 3 & 3 \\ 3 & 2 & 1 \end{vmatrix}=0$<br>$3x-7y+5z+3=0$ | 1<br><br>1<br><br>2<br>1 |
| 46.(a) | Diagram<br>The required Area $= 4\int_0^a y\,dx = 4\int_0^a \frac{b}{a}\sqrt{a^2-x^2}\,dx$<br>$= \frac{4b}{a}\left[\frac{x}{2}\sqrt{a^2-x^2}+\frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right)\right]_0^a$<br>$=\pi ab$ sq.units | 1<br>1<br>2<br>1 |
| 46.(b) | Equation of parabola:$(y-2)^2=8(x-1)$<br>Vertex:(1,2)<br>Focus:(3,2)<br>Equation of directrix: x+1=0 | 2<br>1<br>1<br>1 |



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47.(a)

| p | q | p↔q | ∼p | ∼q | (∼p)∨q | (∼q)∨p | ((∼p)∨q)∧((∼q)∨p) |
|---|---|---|---|---|---|---|---|
| T | T | T | F | F | T | T | T |
| T | F | F | F | T | F | T | F |
| F | T | F | T | F | T | F | F |
| F | F | T | T | T | T | T | T |

4

From column (3),(8)

p↔q≡((∼)p∨q)∧((∼)q∨p) 1

47.(b)

Δ= -15 1

ΔX =-15 1

ΔY = -5 1

ΔZ = -5 1

(X,Y,Z)=(1,3,3) 1

# MARK ANALYSIS

## (WITHOUT CHOICE)

| PART | Questions | Total Questions | Book Back Questions | Interior/Creative Questions | Total Marks |
|---|---|---|---|---|---|
| I | 1 Mark | 20 | 16 | 4 | 20 |
| II | 2 Marks | 10 | 9 | 1 | 20 |
| III | 3 Marks | 10 | 9 | 1 | 30 |
| IV | 5 Marks | 14 | 12 | 2 | 70 |
| Total Marks | | | 121 | 19 | 140 |
| Percentage | | | 86% | 14% | 100% |

# DEPARTMENT OF MATHEMATICS



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